

# SIMATS ENGINEERING

## ASSESSMENT TOOL 1

COURSE CODE: ITA0606

COURSE NAME: MACHINE LEARNING

### COURSE OUTCOME COVERED

**CO1:** *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

*Assessment Tool Weightage: 40%*

QUESTIONS: 15

Total Marks: 25

Annexure A

### APPLICATION BASED QUESTIONS

*Code of Conduct:*

**I Chalamala Sai Harshitha (192424346)** *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*  
*I understand that any violation of academic integrity rules will result in disciplinary action.*

*Signature of the student:* C. Sai Harshitha

1. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C?

**Given**

- **Section A:** 5 Red, 3 Blue, 2 Black
  - Total cars = 10
  - $P(\text{Blue} | A) = \frac{3}{10}$
- **Section B:** 4 Red, 6 Blue, 5 Black
  - Total cars = 15
  - $P(\text{Blue} | B) = \frac{6}{15} = \frac{2}{5}$
- **Section C:** 7 Red, 2 Blue, 6 Black
  - Total cars = 15
  - $P(\text{Blue} | C) = \frac{2}{15}$
- Section **B** is **twice as likely** to be selected as Sections A and C.

Therefore,

$$P(A) = \frac{1}{4}, P(B) = \frac{1}{2}, P(C) = \frac{1}{4}$$

**Formula (Bayes' Theorem)**

$$P(C | \text{Blue}) = \frac{P(\text{Blue} | C) \times P(C)}{P(\text{Blue})}$$

Where;

$$\begin{aligned} P(\text{Blue}) &= P(\text{Blue} | A)P(A) + P(\text{Blue} | B)P(B) + P(\text{Blue} | C)P(C) \\ &= \frac{\left(\frac{3}{10} \times \frac{1}{4}\right)}{\left(\frac{3}{10} \times \frac{1}{4}\right) + \left(\frac{6}{15} \times \frac{1}{2}\right) + \left(\frac{2}{15} \times \frac{1}{4}\right)} \\ &= \frac{\frac{1}{30}}{\frac{3}{40} + \frac{1}{5} + \frac{1}{30}} \end{aligned}$$

Taking LCM = 120,

$$= \frac{\frac{4}{120}}{\frac{9}{120} + \frac{24}{120} + \frac{4}{120}}$$

$$\begin{aligned}
&= \frac{\frac{4}{120}}{\frac{37}{120}} = \frac{4}{120} \times \frac{120}{37} \\
&= \frac{4}{37} \\
&= 0.1081 = 10.81\%
\end{aligned}$$

**Final Answer**

$$P(C \mid \text{Blue}) = \frac{4}{37} = 0.1081 = 10.81\%$$

**Probability = 0.1081 = 10.81%**

2. In a real-world medical image classification task for detecting breast cancer, a deep learning model integrated with the K-Nearest Neighbors algorithm is applied for binary classification (malignant vs. benign). The dataset consists of 1800 samples per class, with 20% of the data reserved for testing. During evaluation, the model reports 210 True Positives (TP), 190 True Negatives (TN), and 30 False Positives (FP). Based on the given information, determine the missing False Negatives (FN) value, and then calculate the Accuracy, Precision, Sensitivity (Recall), and Specificity of the classifier, explaining its effectiveness in medical diagnosis.

**Given**

- Total samples per class = **1800**
- Test data = **20%**
- Positive (Malignant) test samples =  $1800 \times 20\% = 360$
- True Positives (TP) = **210**
- True Negatives (TN) = **190**
- False Positives (FP) = **30**

**Step 1: Find False Negatives (FN)**

Since the positive test samples are **360**,

$$TP + FN = 360$$

$$FN = 360 - 210 = 150$$

$$\boxed{FN = 150}$$

**Step 2: Calculate Performance Metrics**

**1. Accuracy**

$$\begin{aligned}
\text{Accuracy} &= \frac{TP + TN}{TP + TN + FP + FN} \\
&= \frac{210 + 190}{210 + 190 + 30 + 150} = \frac{400}{580} = 0.6897
\end{aligned}$$

$$\boxed{\text{Accuracy} = 68.97\%}$$

## 2. Precision

$$\begin{aligned}\text{Precision} &= \frac{TP}{TP + FP} \\ &= \frac{210}{210 + 30} = \frac{210}{240} = 0.875\end{aligned}$$

$$\boxed{\text{Precision} = 87.50\%}$$

## 3. Sensitivity (Recall)

$$\begin{aligned}\text{Sensitivity} &= \frac{TP}{TP + FN} \\ &= \frac{210}{210 + 150} = \frac{210}{360} = 0.5833\end{aligned}$$

$$\boxed{\text{Sensitivity (Recall)} = 58.33\%}$$

## 4. Specificity

$$\begin{aligned}\text{Specificity} &= \frac{TN}{TN + FP} \\ &= \frac{190}{190 + 30} = \frac{190}{220} = 0.8636\end{aligned}$$

$$\boxed{\text{Specificity} = 86.36\%}$$

## Conclusion

The model has high precision and specificity, meaning it correctly identifies most predicted malignant and healthy cases. However, its low sensitivity (58.33%) indicates that many actual malignant cases are missed, making it less reliable for medical diagnosis where detecting all positive cases is crucial.

### 3. Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs:

- (a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%
- (b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63%
- (c) Epoch 3: Training Accuracy = 70%, Validation Accuracy = 68%
- (d) Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72%
- (e) Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73%
- (f) Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72%
- (g) Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70%
- (h) Epoch 8: Training Accuracy = 92%, Validation Accuracy = 68%
- i) Identify the epoch at which the model starts overfitting.

ii) Explain why validation accuracy decreases despite training accuracy increasing.

iii) Suggest two methods to improve validation performance.

Given

| Epoch | Training Accuracy | Validation Accuracy |
|-------|-------------------|---------------------|
| 1     | 60%               | 58%                 |
| 2     | 65%               | 63%                 |
| 3     | 70%               | 68%                 |
| 4     | 75%               | 72%                 |
| 5     | 80%               | 73%                 |
| 6     | 85%               | 72%                 |
| 7     | 88%               | 70%                 |
| 8     | 92%               | 68%                 |

i) Identify the epoch at which the model starts overfitting.

The model starts **overfitting at Epoch 6**.

Reason:

- Up to **Epoch 5**, both training and validation accuracy improve.
- From **Epoch 6** onward, the **training accuracy continues to increase**, but the **validation accuracy begins to decrease**.
- This indicates that the model is learning the training data too specifically and is not generalizing well to unseen data.

Answer:

|                                          |
|------------------------------------------|
| The model starts overfitting at Epoch 6. |
|------------------------------------------|

ii) Explain why validation accuracy decreases despite training accuracy increasing.

As training continues, the model begins to **memorize the training data**, including noise and unimportant patterns, instead of learning general features.

Therefore:

- **Training accuracy increases** because the model fits the training data more closely.
- **Validation accuracy decreases** because the model performs poorly on new, unseen data.

This is known as **overfitting**, where the model has **high training performance but low generalization ability**.

iii) Suggest two methods to improve validation performance.

### 1. Early Stopping

Stop training when the validation accuracy stops improving. This prevents the model from learning noise and reduces overfitting.

## 2. Dropout Regularization

Randomly deactivate some neurons during training. This forces the model to learn more robust features and improves its ability to generalize to unseen data.

### 4. In a spam email filtering system, an AI model classifies emails as Spam or Not Spam.

Out of 400 emails (200 spam and 200 non-spam):

- Spam identified as Spam: 180
- Spam identified as Not Spam: 20
- Not Spam identified as Not Spam: 170
- Not Spam identified as Spam: 30

Compute: Accuracy, Precision, Sensitivity, Specificity, and F1-Score.

Given

- Total emails = 400
- Spam emails = 200
- Not Spam emails = 200

Confusion Matrix:

- **True Positive (TP)** = Spam identified as Spam = 180
- **False Negative (FN)** = Spam identified as Not Spam = 20
- **True Negative (TN)** = Not Spam identified as Not Spam = 170
- **False Positive (FP)** = Not Spam identified as Spam = 30

#### 1. Accuracy

**Formula**

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \times 100$$

**Calculation**

$$\begin{aligned} &= \frac{180 + 170}{180 + 170 + 30 + 20} \times 100 \\ &= \frac{350}{400} \times 100 \\ &= 87.5\% \end{aligned}$$

**Accuracy = 87.5%**

#### 2. Precision

**Formula**

$$\text{Precision} = \frac{TP}{TP + FP} \times 100$$

### Calculation

$$\begin{aligned} &= \frac{180}{180 + 30} \times 100 \\ &= \frac{180}{210} \times 100 \\ &= 85.71\% \end{aligned}$$

**Precision = 85.71%**

### 3. Sensitivity (Recall)

#### Formula

$$\text{Sensitivity (Recall)} = \frac{TP}{TP + FN} \times 100$$

#### Calculation

$$\begin{aligned} &= \frac{180}{180 + 20} \times 100 \\ &= \frac{180}{200} \times 100 \\ &= 90\% \end{aligned}$$

**Sensitivity (Recall) = 90%**

### 4. Specificity

#### Formula

$$\text{Specificity} = \frac{TN}{TN + FP} \times 100$$

#### Calculation

$$\begin{aligned} &= \frac{170}{170 + 30} \times 100 \\ &= \frac{170}{200} \times 100 \\ &= 85\% \end{aligned}$$

**Specificity = 85%**

### 5. F1-Score

#### Formula

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

### Calculation

Using decimal values:

$$\begin{aligned} &= \frac{2 \times 0.8571 \times 0.90}{0.8571 + 0.90} \\ &= \frac{1.5428}{1.7571} \\ &= 0.8780 \\ &= 87.80\% \end{aligned}$$

**F1-Score = 87.80%**

5. A warehouse has two sections. Section A contains 15 defective items and 35 non defective items, while Section B contains 25 defective items and 25 non-defective items. One section is chosen at random, but Section A is three times as likely to be chosen as Section B. An item selected from the chosen section is found to be defective.

### Given

- **Section A**
  - Defective items = 15
  - Non-defective items = 35
  - Total items = 50
- **Section B**
  - Defective items = 25
  - Non-defective items = 25
  - Total items = 50
- Section A is **three times as likely** to be chosen as Section B.

Therefore,

$$P(A) = \frac{3}{4}, P(B) = \frac{1}{4}$$

The selected item is **defective**.

Find:

$$P(B \mid \text{Defective})$$

### Apply Bayes' Theorem

$$P(B \mid \text{Defective}) = \frac{P(\text{Defective} \mid B) \times P(B)}{P(\text{Defective} \mid A) \times P(A) + P(\text{Defective} \mid B) \times P(B)}$$

$$= \frac{\left(\frac{25}{50} \times \frac{1}{4}\right)}{\left(\frac{15}{50} \times \frac{3}{4}\right) + \left(\frac{25}{50} \times \frac{1}{4}\right)}$$

$$= \frac{\left(\frac{1}{2} \times \frac{1}{4}\right)}{\left(\frac{3}{10} \times \frac{3}{4}\right) + \left(\frac{1}{2} \times \frac{1}{4}\right)}$$

$$= \frac{\frac{1}{8}}{\frac{9}{40} + \frac{1}{8}}$$

$$= \frac{\frac{1}{8}}{\frac{9}{40} + \frac{5}{40}}$$

$$= \frac{\frac{1}{8}}{\frac{14}{40}}$$

$$= \frac{1}{8} \times \frac{40}{14} = \frac{5}{14}$$

$$= 0.3571 = 35.71\%$$

### Final Answer

|                                                                |
|----------------------------------------------------------------|
| $P(B \mid \text{Defective}) = \frac{5}{14} = 0.3571 = 35.71\%$ |
|----------------------------------------------------------------|

## RUBRICS FOR CALCULATION

| Criteria                          | Weightage | Level 4 – Exemplary (5 Marks)                                | Level 3 – Proficient (4 Marks)         | Level 2 – Developing (2–3 Marks)             | Level 1 – Beginning (0–1 Mark)  |
|-----------------------------------|-----------|--------------------------------------------------------------|----------------------------------------|----------------------------------------------|---------------------------------|
| <b>Conceptual Understanding</b>   | 25        | Demonstrates clear and complete understanding of the concept | Good understanding with minor gaps     | Basic understanding with some misconceptions | Poor or incorrect understanding |
| <b>Accuracy of Answer</b>         | 25        | Completely correct answer with no errors                     | Mostly correct with minor errors       | Partially correct with noticeable errors     | Incorrect or irrelevant answer  |
| <b>Application of Knowledge</b>   | 20        | Correctly applies concepts to solve the question/problem     | Applies concepts with minor mistakes   | Limited or partially correct application     | No or incorrect application     |
| <b>Completeness of Response</b>   | 15        | Covers all required parts of the question                    | Covers most parts with minor omissions | Covers only some parts                       | Incomplete or missing response  |
| <b>Clarity &amp; Presentation</b> | 15        | Clear, concise, and well-organized answer                    | Understandable with minor issues       | Somewhat unclear or poorly structured        | Difficult to understand         |